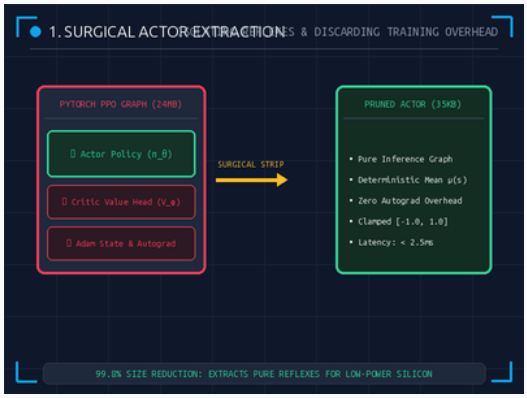
■ MICRODUCK PHYSICAL AI MASTERCLASS

Module 4: Brain Surgery: Actor Isolation, Hardware Clamps, ONNX Export & Temporal Memory

PHASE 4 HANDOUT

Edge Inference

PRUNED SIZE: 35 KB (99.8% Drop)	INFERENCE: 2.5ms on RK3566	SAFETY: torch.clamp()	BUFFER: deque(maxlen=4)



■ 1. Surgical Actor Extraction

REFLEX ISOLATION & 99.8% PRUNING

Pruning Training Overhead: The 24MB PyTorch checkpoint contains the Critic value network $V_\phi(s)$, Adam optimizer state, and backward autograd graphs. We extract only the 35KB forward Actor $\pi_\theta(a|s)$.

The 12-Year-Old Analogy: When you take a piano exam, your teacher doesn't sit on your lap. You leave the teacher at home and only bring your muscle memory.

- **Actor Model**: Extracts deterministic mean $\mu(s)$ policy graph for direct joint actuation.
- **Weight Freezing**: Discards gradient tracking (`requires_grad=False`), eliminating dynamic memory allocations.
- **Silicon Size**: Reduces model footprint from 24,000 KB down to a lightweight 35 KB binary.

■ 2. Hardware Safety Clamping in Silicon

TORCH.CLAMP() BOUNDS & GEAR PROTECTION

Mathematical Inviolability: Unbounded neural outputs (>1.0) can strip physical nylon servo gears. By baking `torch.clamp(a, -1.0, 1.0)` into the computational graph, motor safety is guaranteed in silicon.

The 12-Year-Old Analogy: Bumper rails on a bowling lane that physically prevent the ball from bouncing into the next lane, no matter how hard you throw it.

- **Silicon Clamping**: Compiled as a native ONNX `Clip` operator executed directly on the NPU.
- **Range Guarantee**: Outputs are mathematically bounded to $[-1.0, 1.0]$ before reaching the PWM bus.
- **Failsafe Redundancy**: Prevents runaway policy gradients from destroying $399 physical hardware.

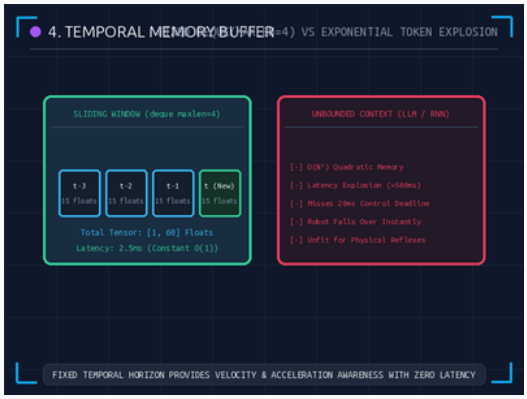
■ 3. ONNX Computational Graph

CROSS-PLATFORM IR & 2.5MS EXECUTION

Standardized Tensor Graph: Compiles PyTorch weights into an ONNX Opset 17 computational DAG (Gemm $\to$ ReLU $\to$ Clip), executable via `onnxruntime` across any CPU/NPU hardware without Python.

The 12-Year-Old Analogy: Translating an English recipe into universal cooking symbols that any chef in any country can read and execute instantly.

- **Execution Speed**: Runs full 60-float inference in <2.5ms on Rockchip RK3566 quad-core ARM.
- **Deterministic Pipeline**: Eliminates Python Global Interpreter Lock (GIL) and runtime garbage collection.
- **Opset 17 Standard**: Portable representation deployable across C++, Rust, or embedded Linux kernels.



■ 4. Temporal Memory & Sliding Buffer

FIXED DEQUE(MAXLEN=4) VS TOKEN EXPLOSION

Fixed Memory Horizon: Balancing requires velocity and acceleration awareness. A fixed sliding window `deque(maxlen=4)` buffers 4 frames $\times$ 15 sensors = 60 floats with constant $O(1)$ memory.

The 12-Year-Old Analogy: Unlike ChatGPT which grows huge and slower the more you talk, the duck's memory is a 4-frame conveyor belt where old frames quietly drop off.

- **Constant O(1) Memory**: Prevents LLM-style $O(N^2)$ quadratic token memory growth and latency spikes.
- **State Dimensions**: Produces flat $[1, 60]$ input tensor capturing instantaneous joint momentum.
- **Strict 50Hz Budget**: 2.5ms inference + 17.5ms cadence sleep = perfectly steady 20.0ms heartbeat.

■ PHASE 4 SILICON DEPLOYMENT SUMMARY & RUST DAEMON HANDOFF

- **Reflex Model:** Extracted 35KB ONNX policy containing pure clamped walking muscle memory.
- **Real-Time Cadence:** Predictable 2.5ms inference guarantees zero timing jitter on edge hardware.
- **Next Module:** Deploy the ONNX binary into Phase 5: The Rust Nervous System (`robotd` & `robotctl`).

■ PHASE 4 KNOWLEDGE CHECK

ASSESSMENT

Test your understanding of Model Extraction, Clamping, ONNX Compilation, and Temporal Memory.

5 Questions

■ Q1: Actor Policy Extraction

MULTIPLE CHOICE

Why do we extract only the Actor network and discard the Critic?

- **A.** Because the Critic is written in a different programming language.
- **B.** The Critic is only needed to score actions during training; the Actor contains all walking reflexes.
- **C.** The Critic network causes the robot to walk backwards.

■ Q2: Silicon Clamping Protection

MULTIPLE CHOICE

What is the primary danger of deploying an ONNX policy without torch.clamp()?

- **A.** Runaway neural outputs (>1.0) can command excessive speed, stripping physical gears.
- **B.** The robot's battery will charge too quickly.
- **C.** The WiFi antenna will disconnect.

■ Q3: ONNX Edge Execution

MULTIPLE CHOICE

What is the advantage of compiling PyTorch models to ONNX for edge robotics?

- **A.** ONNX allows the robot to fly.
- **B.** ONNX represents pure math graphs, executing in <2.5ms without heavy Python overhead.
- **C.** ONNX turns the camera into an IMU.

■ Q4: 50Hz Timing Discipline

MULTIPLE CHOICE

Why does our 50Hz control loop explicitly sleep for 17.5ms after a 2.5ms inference?

- **A.** To maintain a perfectly steady 20ms heartbeat and prevent motor timing jitter.
- **B.** To allow the CPU to cool down to 0 degrees.
- **C.** Because Linux cannot run for more than 3ms.

■ Q5: Sliding Temporal Memory

MULTIPLE CHOICE

Why does deque(maxlen=4) avoid the 'token explosion' problem of LLMs?

- **A.** It deletes the robot's brain every second.
- **B.** It holds a fixed 60-float buffer, ensuring constant O(1) memory and predictable 2.5ms latency.
- **C.** It compresses video into text files.

Answer Key: 1-B, 2-A, 3-B, 4-A, 5-B (1-B: Actor reflexes; 2-A: Gear strip prevention; 3-B: Fast math graph; 4-A: Precise 20ms cadence; 5-B: Constant O(1) buffer)